

Browser Audio Lab

ToolsPrivacy

SIGNAL & FREQUENCY

Tone Generator

Generate a controlled digital oscillator, switch waveform and stereo channel live, and tune frequency with a logarithmic control.

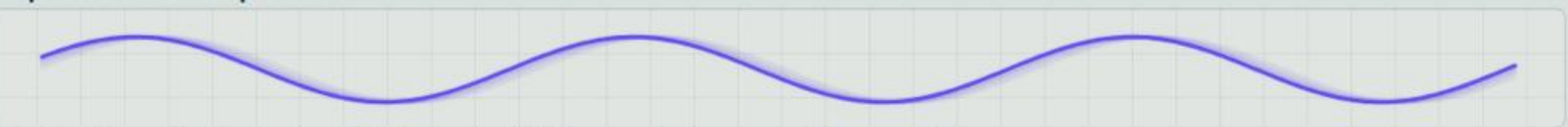
● Playing

REQUESTED FREQUENCY
440 Hz

20 Hz-20 kHz · digital request

GENERATED SIGNAL

Requested waveform preview



Stylized waveform preview. The numeric control is the requested digital frequency; this is not an acoustic measurement.

FREQUENCY

Frequency

440

401004401k10k

SIGNAL

Waveform

SineSquareTriangleSaw

Output

LeftBothRight

LEVEL

Level

-24 dB

dB relative to unity gain

TRANSPORT

Stop

Short click-reduction fade on stop.

LOW VOLUME

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.

WHAT THIS TOOL DOES

A direct browser-

The displayed frequency is the requested digital oscillator frequency generated through the Web Audio API. It is useful for checking channels, comparing tones

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SIGNAL & FREQUENCY

Bass Test

Explore low-frequency playback with a selected tone, a slow logarithmic sweep and a stepped preset sequence while listening for rattles or resonances.

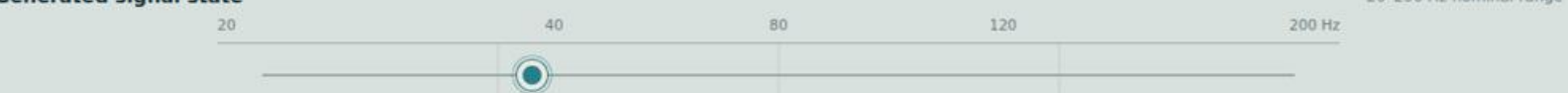
● Slow bass sweep running

GENERATED FREQUENCY
20 Hz

20 Hz · sweep

LOW-FREQUENCY REFERENCE

Generated signal state



Reference visualization only. The numeric readout is the requested/generated digital frequency; it is not measured speaker motion, acoustic output, or frequency response.

MODE

Single toneSlow sweepPreset sequence

20 → 120 Hz

12 seconds · logarithmic

Slowly move through the core low-frequency range to listen for rattles or resonant spots.

Run slow sweep

OUTPUT

Level

-24 dB

dB relative to unity gain

Stop

Low volume

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Keep playback volume moderate. If you cannot hear a low tone, do not turn the system volume up to compensate.

WHAT THIS CHECKS

Bass Test generates controlled digital sine tones from a nominal 20-200 Hz range,

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OUTPUT DIAGNOSTICS

Sound Test

Run a fast output smoke test with the same short reference signal on the left channel, both channels or the right channel.

● Playing Left

REFERENCE BURST
500 Hz

700 ms sine

DIGITAL CHANNEL TARGET

Hard-routed stereo reference



Active channel: **Left**. This is the requested digital route; the browser cannot verify physical speaker output.

SINGLE BURST

LeftBothRight

GUIDED SEQUENCE

Left → Both → Right

700 ms burst · 300 ms gap

Run sequence

TRANSPORT

Same source

500 Hz in every channel step

Stop

Low volume

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Use your ears to confirm the expected channel; Browser Audio Lab does not detect which physical speaker produced sound.

WHAT TO LISTEN FOR

Confirm the

Left, Both and Right each use the same browser-generated 500 Hz sine burst. Use them as a quick check that audio is reaching the output you expect before moving

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SIGNAL & FREQUENCY

Frequency Sweep

Configure a controlled digital sweep, then listen across the selected range.

● Frequency sweep running

20 Hz → 20 kHz

15 s · Logarithmic

SCHEDULED FREQUENCY PATH

Requested digital sweep



Stylized schedule visual. Endpoints, duration and cursor describe the requested/generated digital sweep; this page does not measure acoustic frequency response.

RANGE

Low frequencyHigh frequency

2020000

TIMING & PATH

Duration

15 s

5 s60 s

Scale

LinearLogarithmicAscendingDescending

Direction

Level

-24 dB

dB relative to unity gain

TRANSPORT

Play sweep

Stop

LOW VOLUME

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.

WHAT THIS DOES

Frequency Sweep generates one digital oscillator across the requested frequency

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SIGNAL & FREQUENCY

Noise Generator

Play deterministic white, pink or brown reference noise locally, with a controlled digital Level and optional timer.

● White noise · Timer off

GENERATED REFERENCE
White noise

Timer off

LOCAL REFERENCE BUFFER

Noise source state



Reference visualization only. It does not show a spectrum, acoustic measurement, loudness or laboratory spectral analysis.

NOISE TYPE

WhitePinkBrown

TIMER

Off1 min5 min10 min

OUTPUT

Level

-24 dB

dB relative to unity gain

Play noise

Stop

Comfort level

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Generated digital noise only; Browser Audio Lab cannot measure the physical loudness reaching your ears.

REFERENCE SIGNAL

Noise Generator creates reusable reference-noise PCM at the canonical 44,100 Hz

tone-generator

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Tools Privacy

SIGNAL & FREQUENCY

Tone Generator

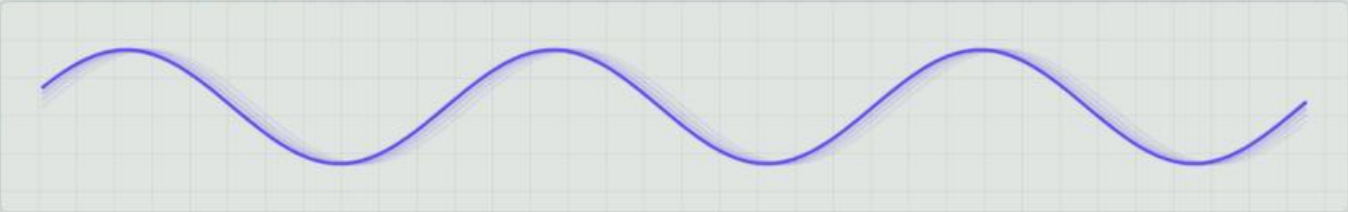
Generate a controlled digital oscillator, switch waveform and stereo channel live, and tune frequency with a logarithmic control.

● Playing

REQUESTED FREQUENCY
440 Hz

20 Hz-20 kHz · digital request

GENERATED SIGNAL
Requested waveform preview



Stylized waveform preview. The numeric control is the requested digital frequency; this is not an acoustic measurement.

FREQUENCY
Frequency

440

Hz

401004401k10k

SIGNAL
Waveform

Sine

Square

Triangle

Saw

Output

LeftBothRight

LEVEL
Level

-24 dB

dB relative to unity gain

TRANSPORT

Stop

Short click-reduction fade on stop.

LOW VOLUME

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.

bass-test

🚩 Browser Audio Lab

Tools Privacy

SIGNAL & FREQUENCY

Bass Test

Explore low-frequency playback with a selected tone, a slow logarithmic sweep and a stepped preset sequence while listening for rattles or resonances.

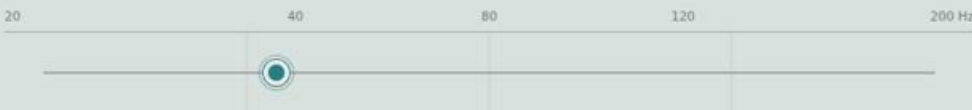
● Slow bass sweep running

GENERATED FREQUENCY
20 Hz

20 Hz · sweep

LOW-FREQUENCY REFERENCE
Generated signal state

20-200 Hz nominal range



Reference visualization only. The numeric readout is the requested/generated digital frequency; it is not measured speaker motion, acoustic output, or frequency response.

MODE

Single tone

Slow sweep

Preset sequence

20 → 120 Hz

12 seconds · logarithmic

Slowly move through the core low-frequency range to listen for rattles or resonant spots.

Run slow sweep

OUTPUT
Level

-24 dB

dB relative to unity gain

Stop

Low volume

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Keep playback volume moderate. If you cannot hear a low tone, do not turn the system volume up to compensate.

WHAT THIS CHECKS

Low-frequency

Bass Test generates controlled digital sine tones from a nominal 20-200 Hz range, capped when the browser audio sample rate requires it. Use Single tone for one selected frequency. Slow sweep for the nominal 20-120 Hz reference sweep, or

sound-test

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Tools Privacy

OUTPUT DIAGNOSTICS

Sound Test

Run a fast output smoke test with the same short reference signal on the left channel, both channels or the right channel.

● Playing Left

REFERENCE BURST
500 Hz

700 ms sine

DIGITAL CHANNEL TARGET
Hard-routed stereo reference

Left · Both · Right



Active channel: Left. This is the requested digital route; the browser cannot verify physical speaker output.

SINGLE BURST

LeftBothRight

GUIDED SEQUENCE
Left → Both → Right

700 ms burst · 300 ms gap

Run sequence

TRANSPORT
Same source

500 Hz in every channel step

Stop

Low volume

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Use your ears to confirm the expected channel; Browser Audio Lab does not detect which physical speaker produced sound.

WHAT TO LISTEN FOR

Confirm the expected channel

Left, Both and Right each use the same browser-generated 500 Hz sine burst. Use them as a quick check that audio is reaching the output you expect before moving on to more detailed diagnostics.

frequency-sweep

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Tools Privacy

SIGNAL & FREQUENCY

Frequency Sweep

Configure a controlled digital sweep, then listen across the selected range.

● Frequency sweep running

20 Hz → 20 kHz

15 s · Logarithmic

SCHEDULED FREQUENCY PATH
Requested digital sweep

20 Hz-20 kHz reference scale



Stylized schedule visual. Endpoints, duration and cursor describe the requested/generated digital sweep; this page does not measure acoustic frequency response.

RANGE
Low frequency

20

High frequency

20000

TIMING & PATH
Duration

15 s

5 s60 s

Scale

Linear

Logarithmic

Direction

Ascending

Descending

LEVEL
Level

-24 dB

dB relative to unity gain

TRANSPORT

Play sweep

Stop

LOW VOLUME

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.

WHAT THIS DOES

One scheduled sweep, controlled

Frequency Sweep generates one digital oscillator across the requested frequency interval. Choose a low and high endpoint, a 5-60 second duration, linear or logarithmic progression, and Ascending or Descending direction. The browser

noise-generator

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Tools Privacy

SIGNAL & FREQUENCY

Noise Generator

Play deterministic white, pink or brown reference noise locally, with a controlled digital Level and optional timer.

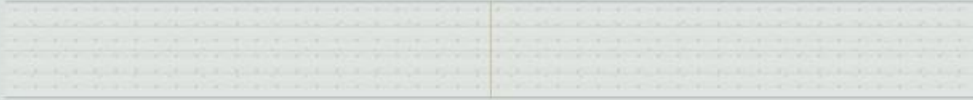
● White noise · Timer off

GENERATED REFERENCE
White noise

Timer off

LOCAL REFERENCE BUFFER
Noise source state

44.1 kHz · 8 s loop · mono source to L/R



Reference visualization only. It does not show a spectrum, acoustic measurement, loudness or laboratory spectral analysis.

NOISE TYPE

White

Pink

Brown

TIMER

Off

1 min

5 min

10 min

OUTPUT
Level

-24 dB

dB relative to unity gain

Play noise

Stop

Comfort level

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Generated digital noise only; Browser Audio Lab cannot measure the physical loudness reaching your ears.

REFERENCE SIGNAL

Deterministic local

Noise Generator creates reusable reference-noise PCM at the canonical 44,100 Hz rate. White, pink and brown modes use the repository's fixed deterministic algorithms and seeds, normalize source peak to at most 0.8, and condition the

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SIGNAL & FREQUENCY

Tone Generator

Generate a controlled digital oscillator, switch waveform and stereo channel live, and tune frequency with a logarithmic control.

● Playing

REQUESTED FREQUENCY440 Hz

GENERATED SIGNAL

Requested waveform preview



Stylized waveform preview. The numeric control is the requested digital frequency; this is not an acoustic measurement.

FREQUENCY

Frequency

440

401004401k10k

Hz

SIGNAL

Waveform

SineSquareTriangleSaw

Output

LeftBothRight

LEVEL

Level

-24 dB

dB relative to unity gain

TRANSPORT

Stop

Short click-reduction fade on stop.

LOW VOLUMEStart with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.

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ToolsPrivacy

SIGNAL & FREQUENCY

Bass Test

Explore low-frequency playback with a selected tone, a slow logarithmic sweep and a stepped preset sequence while listening for rattles or resonances.

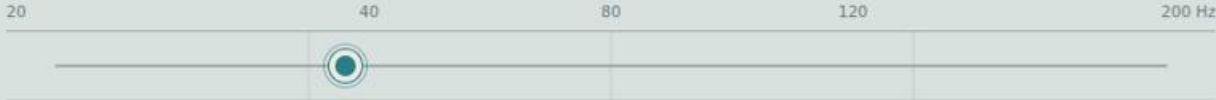
● Slow bass sweep running

GENERATED FREQUENCY20 Hz

20 Hz · sweep

LOW-FREQUENCY REFERENCE

Generated signal state



Reference visualization only. The numeric readout is the requested/generated digital frequency; it is not measured speaker motion, acoustic output, or frequency response.

MODE

Single tone

Slow sweep

Preset sequence

20 → 120 Hz

12 seconds · logarithmic

Slowly move through the core low-frequency range to listen for rattles or resonant spots.

Run slow sweep

OUTPUT

Level

-24 dB

dB relative to unity gain

Stop

Low volumeStart with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Keep playback volume moderate. If you cannot hear a low tone, do not turn the system volume up to compensate.

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OUTPUT DIAGNOSTICS

Sound Test

Run a fast output smoke test with the same short reference signal on the left channel, both channels or the right channel.

● Playing Left

REFERENCE BURST500 Hz

700 ms sine

DIGITAL CHANNEL TARGET

Hard-routed stereo reference



Active channel: Left. This is the requested digital route; the browser cannot verify physical speaker output.

SINGLE BURST

LeftBothRight

GUIDED SEQUENCE

Left → Both → Right

700 ms burst · 300 ms gap

Run sequence

TRANSPORT

Same source

500 Hz in every channel step

Stop

Low volumeStart with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Use your ears to confirm the expected channel; Browser Audio Lab does not detect which physical speaker produced sound.

WHAT TO LISTEN FORLeft: Both and Right each use the same browser-generated 500 Hz sine burst. Use

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SIGNAL & FREQUENCY

Frequency Sweep

Configure a controlled digital sweep, then listen across the selected range.

● Frequency sweep running

20 Hz → 20 kHz

15 s · Logarithmic

SCHEDULED FREQUENCY PATH

Requested digital sweep



Stylized schedule visual. Endpoints, duration and cursor describe the requested/generated digital sweep; this page does not measure acoustic frequency response.

RANGE

Low frequency

High frequency

20

20000

Duration

15 s

5 s60 s

Scale

LinearLogarithmic

Direction

AscendingDescending

LEVEL

Level

-24 dB

dB relative to unity gain

TRANSPORT

Play sweep

Stop

LOW VOLUMEStart with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.

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SIGNAL & FREQUENCY

Noise Generator

Play deterministic white, pink or brown reference noise locally, with a controlled digital Level and optional timer.

● White noise · Timer off

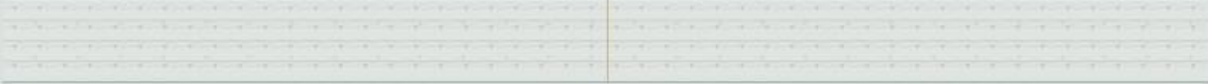
GENERATED REFERENCE

White noise

Timer off

LOCAL REFERENCE BUFFER

Noise source state



Reference visualization only. It does not show a spectrum, acoustic measurement, loudness or laboratory spectral analysis.

NOISE TYPE

WhitePinkBrown

TIMER

Off1 min5 min10 min

OUTPUT

Level

-24 dB

dB relative to unity gain

Play noise

Stop

Comfort levelStart with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear. Generated digital noise only; Browser Audio Lab cannot measure the physical loudness reaching your ears.

tone-generator

Waveform

Sine

Square

Triangle

Saw

Output

Left

Both

Right

LEVEL

Level

-24 dB

dB relative to unity gain

TRANSPORT

Stop

Short click-reduction fade on stop.

LOW VOLUME

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.

WHAT THIS TOOL DOES

A direct browser-generated reference tone

The displayed frequency is the requested

frequency-sweep

Duration

15 s

Scale

Linear

Logarithmic

Direction

Ascending

Descending

LEVEL

Level

-24 dB

dB relative to unity gain

TRANSPORT

Play sweep

Stop

LOW VOLUME

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.

WHAT THIS DOES

One scheduled sweep, controlled by you

Frequency Sweep generates one digital oscillator across the requested frequency

bass-test

Browser Audio Lab

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SIGNAL & FREQUENCY

Bass Test

Explore low-frequency playback with a selected tone, a slow logarithmic sweep and a stepped preset sequence while listening for rattles or resonances.

Slow bass sweep running

20 Hz

20 Hz · sweep

LOW-FREQUENCY REFERENCE

Generated signal state

20

40

80

120

200 Hz

Reference visualization only. The numeric readout is the requested/generated digital frequency; it is not measured speaker motion, acoustic output, or frequency response.

MODE

Single tone

Slow sweep

Preset sequence

20 → 120 Hz

12 seconds · logarithmic

Slowly move through the core low-frequency range to listen for rattles or resonant spots.

Run slow sweep

OUTPUT

Level

-24 dB

dB relative to unity gain

noise-generator

Browser Audio Lab

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SIGNAL & FREQUENCY

Noise Generator

Play deterministic white, pink or brown reference noise locally, with a controlled digital Level and optional timer.

White noise · Timer off

White noise

Timer off

LOCAL REFERENCE BUFFER

Noise source state

Reference visualization only. It does not show a spectrum, acoustic measurement, loudness or laboratory spectral analysis.

NOISE TYPE

White

Pink

Brown

TIMER

Off

1 min

5 min

10 min

OUTPUT

Level

-24 dB

dB relative to unity gain

Play noise

Stop

sound-test

Browser Audio Lab

Tools

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OUTPUT DIAGNOSTICS

Sound Test

Run a fast output smoke test with the same short reference signal on the left channel, both channels or the right channel.

Playing Left

500 Hz

700 ms sine

DIGITAL CHANNEL TARGET

Hard-routed stereo reference

L LEFT

~

R RIGHT

Active channel: **Left**. This is the requested digital route; the browser cannot verify physical speaker output.

SINGLE BURST

Left

Both

Right

GUIDED SEQUENCE

Left → Both → Right

700 ms burst · 300 ms gap

Run sequence

TRANSPORT

Same source

500 Hz in every channel step

Stop

Low volume

Start with your device/headphone volume low. Increase it only to a comfortable listening level. Do not turn the volume up to compensate for a tone you cannot hear.